

Topology

Stuart Anderson

Introduction

These are notes from a Texas-style topology course that I took from Dr. Stuart Anderson in Spring 1990 at Texas A & M, Commerce. This was the first Texas-style course I had ever taken.

The course is fairly self-contained, except as noted at the beginning. Near the end of the course, Dr. Anderson spent about a week lecturing on the Axiom of Choice, its equivalence to Zorn's Lemma and Zermelo's Theorem, and its relation to Tychonoff's Theorem.

As with many Texas-style courses, the biggest challenge for the instructor is to keep all the students involved. I have observed Texas-style classes where a few students end up proving all the theorems, or where a few students never present any proofs. One way to prevent this is to split the proof of a theorem into several lemmas. This provides some easier proofs for weaker students. If all else fails, the instructor can always ask a student to have a proof ready for the next class.

Finally, a set of notes for a Texas-style course should be used only as an outline for a course. The instructor should feel free to make changes to suit his class. I have never used the same set of notes twice.

Good luck. These are a great set of notes!

James Ochoa

1 Preliminaries

Basic notation and terminology of sets and functions are assumed.

Theorem 1 (DeMorgan's Laws): Let A and B be subsets of the set X . Then

1. $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ and
2. $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$.

Definition 2: Let I be an indexing set. For each $\delta \in I$, let A_δ be a set. We define the following two sets:

1. $\bigcup_{\delta \in I} A_\delta = \{x \mid \text{there exists } \delta \in I \text{ such that } x \in A_\delta\}$ and
2. $\bigcap_{\delta \in I} A_\delta = \{x \mid \text{for all } \delta \in I, x \in A_\delta\}$

Theorem 3 (Generalized DeMorgan's Laws): Let $\{A_\delta \mid \delta \in I\}$ be a collection of subsets of the set X . Then

1. $X \setminus \left(\bigcup_{\delta \in I} A_\delta \right) = \bigcap_{\delta \in I} (X \setminus A_\delta)$ and

$$2. X \setminus \left(\bigcap_{\delta \in I} A_\delta \right) = \bigcup_{\delta \in I} (X \setminus A_\delta)$$

Theorem 4: Let A and B be subsets of the set X . Then

$$A \setminus B = A \cap (X \setminus B)$$

Theorem 5: Let $f : X \rightarrow Y$ be a function, and let A and B be subsets of Y . Then

1. $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$,
2. $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$,
3. $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$, and
4. $f^{-1}(A \setminus B) = f^{-1}(A) \setminus f^{-1}(B)$.

Theorem 6: Let $f : X \rightarrow Y$ be a function. Let A be a subset of X , and let B be a subset of Y . Then

1. $A \subset f^{-1}(f(A))$,
2. $f(f^{-1}(B)) \subset B$, and
3. $f(X) \setminus f(A) \subset f(X \setminus A)$.

2 Theorem Sequence

Definition 7: A topological space (X, τ) is a set X and a family of sets τ satisfying the following three conditions:

1. the empty set $\emptyset$, and X , are members of τ ;
2. if A and B are in τ , then $A \cap B$ is in τ ; and
3. if I is an indexing set, and A_δ is in τ for each δ in I , then $\bigcup_{\delta \in I} A_\delta$ is in τ .

The members of τ are called *open sets* and τ is called the *topology on X* .

Definition 8: Let (X, τ) be a topological space. A subset A of X is called a *closed set* if $X \setminus A$ is open.

Theorem 9: The union of finitely many closed sets is closed. The intersection of an arbitrary family of closed sets is closed.

Theorem 10: For any topological space (X, τ) , the sets $\emptyset$ and X are closed.

Theorem 11: Let A be a subset of X . Then A is open if and only if for each x in A , there is an open set O_x such that x is a member of O_x and O_x is a subset of A .

Definition 12: Let (X, τ) be a topological space, and let A be a subset of X . The *interior* of A , notated $\text{int}(A)$, is the union of all open subsets of A . The *exterior* of A , notated $\text{ext}(A)$, is the union of all open sets not intersecting A .

Theorem 13: The interior and exterior operators satisfy the following:

1. $\text{int}(\emptyset) = \emptyset$ and $\text{ext}(\emptyset) = X$,
2. $\text{int}(X) = X$ and $\text{ext}(X) = \emptyset$,
3. $\text{int}(\text{int}(A)) = \text{int}(A)$,
4. $\text{int}(A \cap B) = \text{int}(A) \cap \text{int}(B)$,
5. $\text{ext}(A \cup B) = \text{ext}(A) \cup \text{ext}(B)$,
6. $\text{int}(A) \subseteq A$ and $\text{ext}(A) \subseteq X \setminus A$, and
7. if $A \subseteq B$, then $\text{int}(A) \subseteq \text{int}(B)$ and $\text{ext}(B) \subseteq \text{ext}(A)$.

Definition 14: Let x be a member of X , and let A be a subset of X . Then x is said to be a *boundary point* of A if every open set containing x intersects both A and $X \setminus A$. The set of all boundary points of A , notated ∂A , is called the *boundary* of A .

Theorem 15: For every subset A of X , the sets $\text{int}(A)$, $\text{ext}(A)$, and ∂A , are mutually disjoint and their union is X . Moreover, $\text{int}(A)$ and $\text{ext}(A)$ are open sets, and ∂A is a closed set.

Theorem 16: A set A is closed if and only if $\partial A \subseteq A$. A set A is open if and only if $\partial A \subseteq X \setminus A$.

Definition 17: The *closure* of a set A , notated $\overline{A}$, is the intersection of all closed sets containing A .

Theorem 18: A set A is closed if and only if $\overline{A} = A$.

Theorem 19: If A is a set, then $\overline{A} = \text{int} A \cup \partial A$.

Theorem 20: The closure operator satisfies the following:

1. $\overline{\emptyset} = \emptyset$ and $\overline{X} = X$;
2. $A \subseteq \overline{A}$;
3. $\overline{\overline{A}} = \overline{A}$;
4. $\overline{A \cup B} = \overline{A} \cup \overline{B}$; and
5. if $A \subseteq B$, then $\overline{A} \subseteq \overline{B}$.

Theorem 21: A point x is in $\overline{A}$ if and only if every open set containing x intersects A .

Definition 22: A point x is a *limit point* (also called *cluster point* or *accumulation point*) of a set A if every open set containing x contains a point of A different from x . The derived set of a set A , notated A' , is the set of all limit points of A .

Theorem 23: A set A is closed if and only if $A' \subseteq A$.

Theorem 24: For any set A , $\overline{A} = A \cup A'$.

Definition 25: Let τ and σ be topologies on X . We say that τ is *finer* (or *larger*) than σ if $\sigma \subseteq \tau$. We say that τ is *coarser* (or *smaller*) than σ if $\tau \subseteq \sigma$. If $\tau \subseteq \sigma$ or $\sigma \subseteq \tau$, then the topologies are said to be *comparable*. Otherwise, they are *not comparable*.

Definition 26: A family $\mathcal{B}$ of subsets of a set X is a base for a topology on X if the following two conditions are satisfied:

1. for each x in X , there is a $B \in \mathcal{B}$ such that $x \in B$; and
2. if A and B are in $\mathcal{B}$ and $x \in A \cap B$, then there is a C in $\mathcal{B}$ such that $x \in C$ and $C \subseteq A \cap B$.

Theorem 27: Let $\mathcal{B}$ be a base for a topology on a set X . Let

$$\tau = \{U \mid U \text{ is the union of members of } \mathcal{B}\}$$

Then τ is a topology on X .

Definition 28: The topology τ defined in Theorem 27 is called the *topology generated by* $\mathcal{B}$.

Theorem 29: The topology generated by the base $\mathcal{A}$ is finer than the topology generated by the base $\mathcal{B}$ if and only if for any $B \in \mathcal{B}$ and any $x \in B$, there exists an $A \in \mathcal{A}$ such that $x \in A$ and $A \subseteq B$.

Theorem 30: A family $\mathcal{B}$ of subsets of X is a base for a given topology τ on X if and only if the following two conditions are true:

1. for each U in τ and $x \in U$, there is a $B \in \mathcal{B}$ such that $x \in B$ and $B \subseteq U$, and
2. $\mathcal{B} \subseteq \tau$.

Problem 31: (double-check this one!) Let X be a set. Prove or disprove the following statements.

1. If I is a set and $\{\tau_\delta \mid \delta \in I\}$ is a collection of topologies on X , then $\bigcap_{\delta \in I} \tau_\delta$ is a topology on X .
2. If τ and σ are topologies on X , then $\tau \cup \sigma$ is a topology on X .
3. If $\{\tau_\delta \mid \delta \in I\}$ is a collection of topologies on X , then there exist unique topologies τ and σ on X such that $\tau \subseteq \tau_\delta \subseteq \sigma$ for all $\delta \in I$.

Definition 32: A *metric space* (X, d) is a set X together with a function $d : X \times X \rightarrow \mathfrak{R}$ which satisfies the following conditions:

1. $d(x, y) \geq 0$ for all $x, y \in X$;
2. $d(x, y) = 0$ if and only if $x = y$;
3. $d(x, y) = d(y, x)$ for all $x, y \in X$; and
4. $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

The function d is called a *metric* on X .

Definition 33: Let (X, d) be a metric space. For $x \in X$ and $r > 0$, the set $B(x, r) = \{y \mid y \in X \text{ and } d(x, y) < r\}$ is called the *r-neighborhood* (*r-ball*) about x .

Theorem 34: Let (X, d) be a metric space. The collection of all sets $B(x, r)$ such that $x \in X$ and $r > 0$, is a base for a topology on X .

Definition 35: The topology generated by r -neighborhoods in Theorem 34 is called the *metric topology* on X generated by d .

Definition 36: A topological space (X, τ) is called *metrizable* if there is a metric d on X such that the metric topology on X generated by d is τ .

Theorem 37: Let (X, τ) be a topological space, and $Y \subseteq X$. Let $\tau_Y = \{Y \cap U \mid U \in \tau\}$. Then τ_Y is a topology on Y .

Definition 38: The topological space (Y, τ_Y) in Theorem 37 is called the *relative* (or *induced*) *topology* on Y . Sets in τ_Y are called *open in Y* or *open relative to Y*. Similar terminology is used for closed sets.

Theorem 39: Let (Y, τ_Y) be a subspace of (X, τ) and $A \subseteq Y$. Then

1. A is τ_Y -closed if and only if $A = Y \cap F$, where F is a τ -closed subset of X ;
2. a member x of Y is a τ_Y -limit point of A if and only if x is a τ -limit point of A ;
3. the τ_Y -closure of A is the intersection of Y and the τ -closure of A ; and
4. the intersection of Y and the τ -interior of A is a subset of the τ_Y -interior of A .

Theorem 40: Let (Y, τ_Y) be a subspace of (X, τ) and $A \subseteq Y$. Then

1. if A is closed in Y and Y is closed in X , then A is closed in X ; and
2. if A is open in Y and Y is open in X , then A is open in X .

Definition 41: A topological space (X, τ) is *connected* if X is not the union of two nonempty disjoint open sets. A subset Y of X is connected if (Y, τ_Y) is connected.

Theorem 42: The space X is connected if and only if the only subsets of X which are both open and closed are $\emptyset$ and X .

Theorem 43: Let $\{A_\delta \mid \delta \in I\}$ be a collection of subsets of the set X . If $A_\alpha \cap A_\beta \neq \emptyset$, then $\bigcup_{\delta \in I} A_\delta$ is connected.

Theorem 44: Let A be a subset of X . If A is connected and $A \subseteq B \subseteq \overline{A}$, then B is connected.

Definition 45: Let A and B be subsets of X . We say that A and B are *separated* if $A \cap \overline{B} = \overline{A} \cap B = \emptyset$.

Theorem 46: If A and B are both closed or both open, then the sets $A \setminus B$ and $B \setminus A$ are separated sets.

Theorem 47: A space X is connected if and only if X is not the union of two nonempty separated sets.

Definition 48: A nonempty subset C of X is said to be a *component* of X if

1. C is connected, and
2. if A is any connected subset of X and $A \cap C \neq \emptyset$, then $A \subseteq C$.

If x is a member of X and C is the component of X such that $x \in C$, then we write $C = C(x)$.

Theorem 49: Let x be a member of X . Then the component $C(x)$ is the union of all connected subsets of X containing x .

Theorem 50: Let (X, τ) be a topological space. Then

1. each component of X is closed, and
2. if A and B are distinct components of X , then A and B are separated.

We will accept the following theorem without proof.

Theorem 51: The components of X form a partition of X into maximal connected subsets.

Definition 52: A space X is *locally connected* if it has a basis consisting of connected sets.

Theorem 53: If X is locally connected, then the components of open sets are open.

Theorem 54: A space X is locally connected if and only if for each x in X and each neighborhood U of x , there exists an open connected set V such that $x \in V$ and $V \subseteq U$.

Definition 55: Let (X, τ) and (Y, σ) be topological spaces, $f : X \rightarrow Y$ be a function, and x be a member of X . We say that f is *continuous* at x if the inverse image of every open set containing $f(x)$ is an open set containing x . That is, f is continuous at x if for each open set V containing $f(x)$, there is an open set U such that $x \in U$ and $f(U) \subseteq V$. We say that the function f is continuous if f is continuous at every point in X .

Theorem 56: Let $f : (X, \tau) \rightarrow (Y, \sigma)$. The following six conditions are equivalent:

1. f is continuous,
2. the inverse image of each open subset of Y is open in X ,
3. the inverse image of each closed subset of Y is closed in X ,
4. the inverse image of each member of a base for σ is open in X ,
5. for every subset A of X , $f(\overline{A}) \subseteq \overline{f(A)}$, and
6. for every subset B of Y , $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$.

Definition 57: A function $F : (X, \tau) \rightarrow (Y, \sigma)$ is a homeomorphism if f is one-to-one (notated $1-1$) and onto, and both f and f^{-1} are continuous. In this case, (X, τ) and (Y, σ) are said to be topologically equivalent. Any property which when possessed by a space is possessed by all homeomorphic images of that space, is called a *topological property* or a *topological invariant*.

Theorem 58: If X and Y are topological spaces and f is a $1-1$ function from X onto Y , then the following are equivalent:

1. f is a homeomorphism;
2. if G is a subset of X , then $f(G)$ is open in Y if and only if G is open in X ;
3. if F is a subset of X , then $f(F)$ is closed in Y if and only if F is closed in X ; and
4. if E is a subset of X , then $f(\overline{E}) = \overline{f(E)}$.

Theorem 59: Let X , Y , and Z be topological spaces $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. If f and g are continuous, then $g \circ f : X \rightarrow Z$ is continuous.

Theorem 60: If A is a subset of X and $f : X \rightarrow Y$ is continuous, then $f|_A : A \rightarrow Y$ is continuous.

Theorem 61: If $X = A \cup B$ where A and B are both open (or both closed) in X , and $f : X \rightarrow Y$ is a function such that both $f|_A$ and $f|_B$ are continuous, then f is continuous.

Theorem 62: The continuous image of an connected set is connected. That is, if X is connected and $f : X \rightarrow Y$ is continuous, then $f(X)$ is connected.

Definition 63:

1. A space (X, τ) is called a T_0 -space if for each pair of distinct members of X , there is an open set U containing one of the members but not the other.
2. A space (X, τ) is called a T_1 -space if for each pair of distinct members x and y of X , there is an open set U containing x but not y .
3. A space (X, τ) is called a *Hausdorff* (T_2) space if for each open pair of members x and y in X , there exist disjoint open sets U and V such that $x \in U$ and $y \in V$.
4. A space (X, τ) is called *regular* if for each closed subset K of X and x in X , there exist disjoint open sets U and V such that $K \subseteq U$ and $x \in V$.
5. A space is called a T_3 -space if it is both regular and T_1 .
6. A space (X, τ) is called *normal* if for each pair E and F of disjoint closed subsets of X , there exist disjoint open sets U and V such that $E \subseteq U$ and $F \subseteq V$.
7. A space is called a T_4 -space if it is both normal and T_1 .

Theorem 64: A space (X, τ) is a T_1 -space if and only if singleton sets are closed.

Theorem 65: If (X, τ) is a Hausdorff space, then

1. each finite set is closed; and
2. x is a limit point of a subset A of X if and only if each open set containing x contains infinitely many members of A .

Definition 66: A function $f : (X, \tau) \rightarrow (Y, \sigma)$ is said to be *open* (respectively closed) if the image of each open (respectively closed) set in X is open (respectively closed) in Y .

Theorem 67: If (X, τ) is Hausdorff and $f : (X, \tau) \rightarrow (Y, \sigma)$ is a closed, one-to-one, and onto, then (Y, σ) is Hausdorff.

Theorem 68: A space (X, τ) is regular if and only if for each x in X and each open set U containing x , there exists an open set V such that $x \in V$ and $\overline{V} \subseteq U$.

Theorem 69: A space (X, τ) is normal if and only if for each closed set K and open set U containing K , there exists an open set V such that $K \subseteq V \subseteq \overline{V} \subseteq U$.

Theorem 70: Every metric space is normal.

Theorem 71 (Urysohn's Lemma): A space X is normal if and only if for each pair of disjoint closed sets A and B in X , there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(a) = 0$ for all $a \in A$ and $f(b) = 1$ for all $b \in B$.

Theorem 72 (Tietze's Extension Theorem): A space X is normal if and only if whenever A is a closed subset of X and there is a continuous function $f : A \rightarrow \mathbb{R}$, there exists a continuous extension of f to all of X ; that is, there is a continuous function $F : X \rightarrow \mathbb{R}$ such that $F|_A = f$. Moreover, if f is bounded, then F may be chosen to be bounded also.

Definition 73: A collection of sets $\Phi = \{A_\delta : \delta \in \mathcal{A}\}$ is called a *covering* (or *cover*) of X if $X \subseteq \bigcup_{\delta \in \mathcal{A}} A_\delta$. Any subcollection of Φ which is also a cover of X is called a *subcover*.

Definition 74: A cover Φ of the space X is called an *open cover* of X if each member of Φ is an open subset of X .

Definition 75 (need to think about this one!): A space (X, τ) is *compact* if every open cover of X has a finite subcover. A subset Y of X is said to be compact if (Y, τ_Y) is compact.

Definition 76: A family of sets $\Phi = \{A_\delta \mid \delta \in \mathcal{A}\}$ has the *finite intersection property* if the intersection of each finite subfamily of Φ is nonempty; that is, if Ψ is a finite subset of Φ , then $\bigcap_{\delta \in \Psi} A_\delta$ is a nonempty set.

Theorem 77: A space X is compact if and only if each family of closed subsets of X which has the finite intersection property has a nonempty intersection; that is, if Φ is a collection of closed subsets of X , then $\bigcap_{\delta \in \Phi} A_\delta$ is a nonempty set.

Theorem 78: The continuous image of a compact set is compact.

Theorem 79: A compact subset of a Hausdorff space is closed.

Theorem 80: Disjoint compact subsets of a Hausdorff space have disjoint neighborhoods. That is, if A and B are disjoint compact subsets of a Hausdorff space, then there exist disjoint open sets U and V such that $A \subseteq U$ and $B \subseteq V$.

Definition 81: If X is compact, Y is Hausdorff, and $f : X \rightarrow Y$ is continuous, then f is a *closed map*.

Theorem 82: A one-to-one continuous function from a compact space onto a Hausdorff space is a homeomorphism.

Definition 83: A family φ of subsets of a set X is a *subbase* for a topology on X if for each x in X , there is an S in φ such that $x \in S$.

Theorem 84: Let φ be a base for a topology on X . Let $\mathcal{B}$ be the set of all finite intersections of members of φ . Then $\mathcal{B}$ is a base for a topology on X .

Definition 85: Let Y be a set and let $\{(X_\alpha, \tau_\alpha) \mid \alpha \in A\}$ be a collection of topological spaces. For each α in A , let f_α be a function from Y into X_α . The smallest topology w on Y such that for all α in A , $f_\alpha : Y \rightarrow X_\alpha$ is continuous, is called the *weak topology* on Y .

Theorem 86: Let Y be a set and let $\{(X_\alpha, \tau_\alpha) \mid \alpha \in A\}$ be a collection of topological spaces. For each α in A , let f_α be a function from Y into X_α . The family $\varphi = \{f_\alpha^{-1}(U) \mid U \in \tau_\alpha \text{ and } \alpha \in A\}$ is a subbase for the weak topology w on Y .

Definition 87:

1. The *Cartesian product* of the collection of sets $\{X_\alpha \mid \alpha \in A\}$ is the set

$$\prod_{\alpha \in A} X_\alpha = \{x : A \rightarrow \bigcup_{\alpha \in A} X_\alpha \mid x(\alpha) \in X_\alpha \text{ for each } \alpha \in A\}$$

2. The set X_α is called the α -th *coordinate space*, and $x(\alpha)$ is called the α -th *coordinate* of x .
3. Let $\beta \in A$. The function $P_\beta : \prod_{\alpha \in A} X_\alpha \rightarrow X_\beta$ defined by $P_\beta(x) = x(\beta)$ is called the β -th *projection function*.
4. The *product topology* on $\prod_{\alpha \in A} X_\alpha$ is the weak topology determined by the functions P_α .

Theorem 88: Each projection function is an open function.

Theorem 89: A product space is connected if and only if each coordinate space is connected.

Theorem 90 (Alexander's Sub-base Theorem): Let X be a topological space, and let φ be a subbase for the topology on X . If every open cover of X by members of φ has a finite subcover, then X is compact.

Theorem 91 (Tychonoff's Theorem): A product space is compact if and only if each coordinate space is compact.